

Rajdeep Gill

EDUCATION

University of Manitoba

Bachelor of Science in Computer Engineering

Winnipeg, MB

Sept. 2021 - May 2026

- GPA: 4.45/4.5
- Relevant Coursework: *Engineering Algorithms, Data structures and Algorithms, Systems Engineering Principles*

EXPERIENCE

Undergraduate Research Student

May 2024 - August 2024

University of Manitoba - Biosensors Research Lab

Winnipeg, MB

- Designed and trained a neural network to predict biological cell viability from cell diffraction patterns, achieving an accuracy of 85%, demonstrating the feasibility of non-invasive cell viability prediction.
- Enhanced particle diffraction centroid detection using computer vision, boosting success rate from 75% to 92%, allowing for better statistical analysis.
- Developed an interactive interface for data analysis, allowing for real time visualization of diffraction patterns.

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Winnipeg, MB

- Employed dimensionality reduction techniques, including t-SNE and principal component analysis to compare and cluster datasets, identifying potential correlations and directing future research directions.
- Utilized parallel processing to reduce run times, enhancing research team productivity.
- Streamlined data collection for a separate lab project by developing a program to interface with an oscilloscope.

PROJECTS

Discord Clone 🐙 | *React, TypeScript, HTML/CSS, Next.js*

- Developed a feature-rich Discord Clone with real-time messaging using web sockets
- Utilized Tailwind CSS to create a visually appealing and responsive user interface
- Implemented API services to handle user authentication, voice and video calls.
- Provided users with extensive control, allowing them to edit and delete messages, create and delete servers.

Microprocessor Communication System | *C, Java*

- Designed and implemented a client-server architecture using TCP/IP stack in C, handling multiple concurrent connections
- Developed a Java-based proxy server to interface between an Android application and the microprocessor server.
- Designed the Android application to allow users to control the LEDs, monitor the temperature and button states of the microprocessor.

Chess Game 🏁 | *Python*

- Created a command line chess game using Python
- Implemented features such as move validation, checkmate detection, and piece promotion
- Currently developing multiplayer functionality to enable real-time play between users

AWARDS

NSERC Undergraduate Student Research Award

2023, 2024

Presidents Scholarship

2021-2024

TECHNICAL SKILLS

Languages: Python, Java, C, JavaScript, HTML/CSS, R

Frameworks: React

Developer Tools: Git, VS Code, Visual Studio, Android Studio, IntelliJ

Other: Microsoft Office Suite